GOKUL R.

Chennai, India | gokulrajasekar324@gmail.com | LinkedIn (https://www.linkedin.com/in/gokul6om4/) | GitHub (https://github.com/Gokzz-glitch) | Portfolio (https://gokzz-glitch.github.io/PORTFOLIO/)

PROFESSIONAL SUMMARY

Driven AI & Machine Learning undergraduate with hands-on expertise in **Computer Vision, Edge AI, and Embedded Systems**. Proven ability to develop real-time AI inference models (YOLOv8, PyTorch) and integrate them with hardware platforms (Raspberry Pi, Arduino). Passionate about building embedded vision solutions and seeking to leverage skills in hardware-software integration, image processing, and deep learning for advanced camera systems at e-con Systems.

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning
Easwari Engineering College (SRM Group), Chennai, India | 2024 – 2028

TECHNICAL SKILLS

- **Computer Vision & AI:** OpenCV, PyTorch, YOLOv8, R-CNN, Object Detection, Edge AI Inference, Data Augmentation.
- **Embedded Systems & Hardware:** Raspberry Pi (Embedded Linux), Arduino (C/C++), Microcontrollers, Servo Actuators, Sensor Interfacing, Edge Computing.
- **Languages & Infrastructure:** Python, C/C++ (Arduino), Java, Git/GitHub, Docker, PostgreSQL, Linux.

PROJECTS & RESEARCH

AI-Assisted Colonoscopy Polyp Detection via Hybrid Deep Learning (https://github.com/Gokzz-glitch) | *PyTorch, OpenCV, YOLOv8, R-CNN* | 2026 (In Progress)

- Developing a hybrid YOLOv8 + R-CNN architecture for real-time polyp detection in colonoscopy video, optimizing for edge deployment and targeting >40 FPS inference at clinically relevant mAP thresholds.
- Engineered temporal consistency filtering across video frames to suppress false positives and stabilize bounding-box predictions in high-motion endoscopic sequences.
- Benchmarking detection latency and precision-recall trade-offs against single-stage detectors; designing the pipeline for seamless integration with clinical endoscopy workstation software.

Distributed Systems & Telemetry Backend (ChakraPulse) (https://github.com/Gokzz-glitch) | *PostgreSQL, Docker, Python* | 2025 – 2026

- Engineered a robust distributed telemetry backend system utilizing Docker for containerized deployment and PostgreSQL for reliable data storage.
- Architected infrastructure for collecting and processing sensor data streams, ensuring seamless hardware-to-cloud communication.

HACKATHONS & ACHIEVEMENTS

- **Chemstart Award — SRM Easwari Privilap (1st Year) (https://gokzz-glitch.github.io/PORTFOLIO/chemstar_certificate.jpg)**: Recognized for outstanding performance and innovation in early engineering coursework.
- **Finalist — TRIGENIX-26 National Hackathon (https://gokzz-glitch.github.io/PORTFOLIO/trigenix_hackathon_certificate.jpg)** (Mar 2026): Secured a top 5 ranking among 300+ national teams by developing innovative, autonomous AI software solutions.
- **Finale Qualifier — Triwizardathon 1.0 (https://gokzz-glitch.github.io/PORTFOLIO/triwizardathon_certificate.jpg)** (Microsoft Learn Student Ambassador, Aug 2025): Engineered a functional robotic arm combining servo actuation, microcontrollers, and integrated C/C++ control software under strict 24-hour hackathon constraints.

CERTIFICATIONS

- Oracle Certified Professional
- Cisco Network Addressing & AI Fundamentals
- **Elite + Gold Certification in Human-Computer Interaction (NPTEL) (https://gokzz-glitch.github.io/PORTFOLIO/nptel_hci_certificate.pdf)**